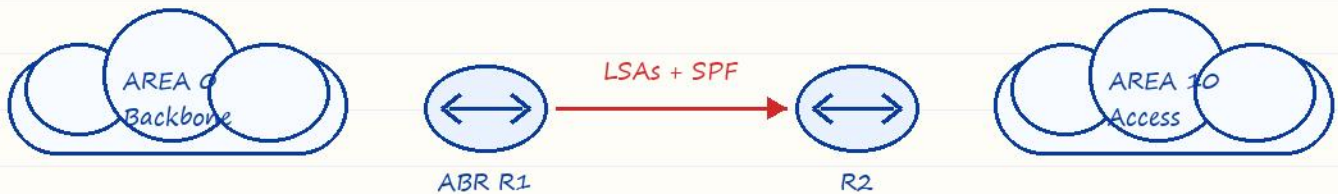


OSPF

(OPEN SHORTEST PATH FIRST)

COMPLETE NOTES



OSPF is a link-state IGP. Routers flood LSAs, build the same LSDB per area, run SPF, then install the best routes.

TOPICS COVERED

- * OSPF fundamentals
- * Link-state database + SPF
- * Packet types and multicast
- * Neighbor states
- * Adjacency requirements
- * DR/BDR and network types
- * LSA types
- * Areas, ABR, ASBR
- * Stub and NSSA areas
- * Metrics and path selection

QUICK HIGHLIGHTS

- * Route summarization
- * Filtering and default routes
- * OSPFv3 for IPv6
- * Authentication and timers
- * Troubleshooting workflow
- * Real-world scenarios
- * Cisco IOS / IOS-XR config
- * Juniper JunOS config
- * Show commands
- * Best practices

WHAT IS OSPF?

- Open Shortest Path First is an Interior Gateway Protocol.
- It uses link-state routing, not hop count.
- Each router builds a topology map for its area.

WHY USE OSPF?

- Fast convergence with triggered updates.
- Hierarchical design with areas.
- Open standard, multi-vendor, widely supported.

WHERE IT RUNS

- Inside one Autonomous System.
- Commonly used in enterprise, service-provider core, campus, and lab networks.
- Carries internal reachability for loopbacks and transit links.

HOW IT THINKS

- Flood LSAs to describe topology.
- Build an LSDB that should match inside the same area.
- Run Dijkstra SPF to select best paths.

KEY TRAITS

- Protocol number 89.
- Administrative distance 110 on Cisco.
- Supports VLSM/CIDR, ECMP, authentication, areas, and summarization.

SUMMARY

- Neighbor up does not guarantee route installed.
- LSDB contents, area type, filtering, and metric decide the final RIB.
- Troubleshoot control plane before data plane.

ROUTE CODES YOU WILL SEE

- O = intra-area route.
- O IA = inter-area route learned through an ABR.
- O E1/E2 = external route from an ASBR.
- O N1/N2 = NSSA external route.

DAILY OPERATOR USE

- Loopbacks for router IDs and iBGP next hops.
- Core transit reachability.
- Campus/collapsed-core routing.
- Fast failure detection when paired with BFD.

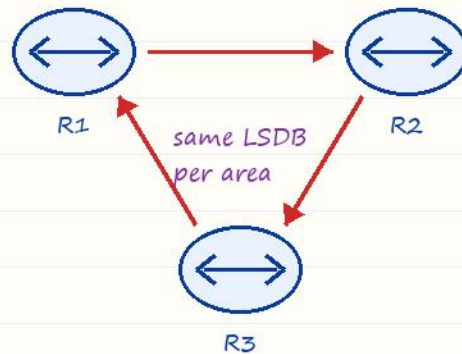
KEY POINT ✳ OSPF is not just 'neighbor up'. The useful chain is interface enabled -> neighbor Full -> LSDB correct -> SPF result -> route installed.

① OSPF vs other routing protocols

Protocol	Model	Best-path input	Operational note
RIP	Distance-vector	Hop count	Simple but slow, max 15 hops
EIGRP	Advanced distance-vector	Composite metric	Cisco-originated, fast convergence
OSPF	Link-state	Cost	Flood LSAs, build LSDB, run SPF
BGP	Path-vector	Policy attributes	Best for inter-AS routing and policy

LINK-STATE IDEA

- Routers describe their own links using LSAs.
- LSAs are flooded reliably to neighbors.
- Every router in an area should calculate from the same LSDB.
- SPF chooses shortest paths by total cost.
- Only changes are flooded after convergence.



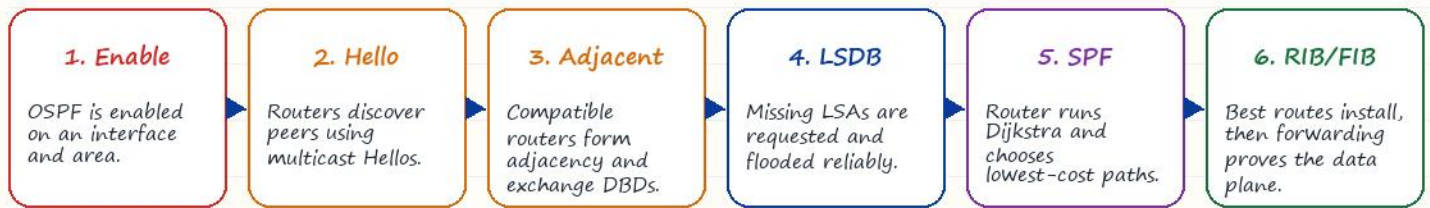
LINK-STATE ADVANTAGE

- Routers converge from a full topology view.
- Triggered updates reduce periodic chatter.
- Areas limit flooding scope and SPF blast radius.

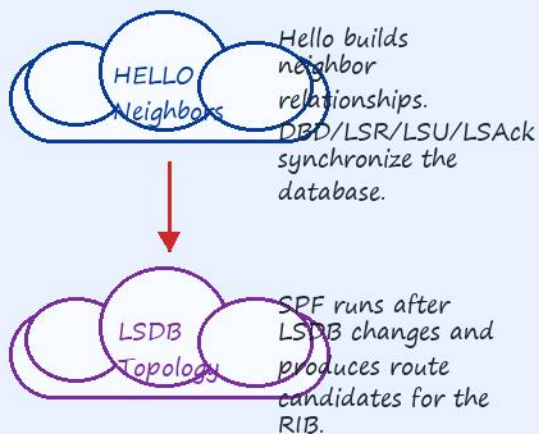
LINK-STATE COST

- More memory/CPU than simple distance-vector protocols.
- Bad LSAs can affect an entire area.
- Design matters: areas, summarization, and stable IDs.

KEY POINT ✨ Distance-vector routers learn rumors from neighbors. OSPF routers flood facts, then each router independently calculates SPF.



OSPF DATABASE FLOW



WHAT CAN BREAK

- No Hello: interface/passive/ACL/problem below IP.
- Hello seen but stuck: timer, area, auth, MTU, or network type.
- Full but no route: area type, LSA missing, filter, summary, or better route.
- Route installed but broken: CEF/FIB, next hop, MTU, policy, or data-plane ACL.

FIRST THREE COMMANDS

```
show ip ospf neighbor
show ip ospf interface brief
show ip ospf database
```

CONTROL PLANE PROVES

- Neighbor state and timers.
- LSA flooding health.
- SPF route decision.

DATA PLANE PROVES

- CEF/FIB next hop.
- Packet path, MTU, and ACL behavior.
- Counters on the exact outgoing interface.

KEY POINT ✨ Work in order: adjacency, database, route table, then forwarding. Jumping straight to packet captures wastes time when OSPF is not even Full.

Packet	Purpose	Destination	Troubleshoot
Hello	Discover/maintain neighbors	224.0.0.5 / 224.0.0.6	Timers, area, auth, stub flag
DBD	Database summary	Neighbor unicast/mcast	Used in ExStart/Exchange
LSR	Request missing LSAs	Neighbor	Seen in Loading
LSU	Send one or more LSAs	Neighbor or multicast	Reliable flooding
LSAck	Acknowledge LSAs	Neighbor or multicast	Stops retransmits

MULTICAST + TRANSPORT

- OSPF uses IP protocol 89, not TCP or UDP.
- 224.0.0.5 = AllSPFRouters.
- 224.0.0.6 = AllDRouters on DR networks.
- Multicast must survive interface ACLs and security filters.
- TTL is 1 for normal OSPF packets.

HELLO PACKET MUST MATCH

- Area ID
- Authentication
- Hello/dead timers
- Stub/NSSA option bits
- Network mask on broadcast/non-broadcast networks
- Compatible network type behavior

PACKET SYMPTOMS

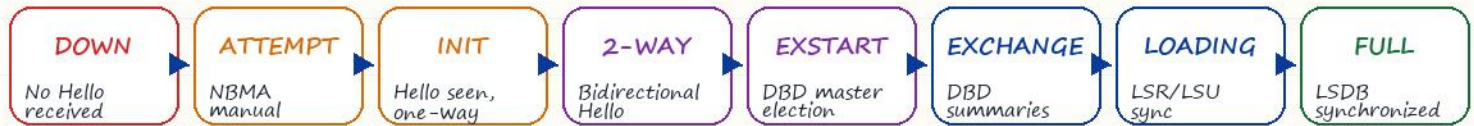
- No Hellos seen: ACL, passive-interface, L2, or wrong subnet.
- DBD retransmits: MTU or unstable link.
- LSR repeats: missing/corrupt LSA or packet loss.

CAPTURE FILTERS

- ip proto 89
- host 224.0.0.5 or host 224.0.0.6
- neighbor source/destination addresses
- Look for DBD MTU and options fields.

KEY POINT ✨ If Hellos do not match, OSPF usually never gets past Init/2-Way. If DBD packets fail, suspect MTU and duplicate router IDs first.

OSPF NEIGHBOR STATES



Dead timer expiry or config mismatch sends the relationship back to DOWN.

State	Meaning	Most Useful Check
Down	No Hello from peer	Check interface, area, ACL, passive-interface
Init	One-way Hello	Check asymmetric ACL/filtering and timers
2-Way	DR election point	Normal for DROTHER pairs on broadcast media
ExStart	DBD negotiation	MTU mismatch or duplicate router ID
Exchange	DBD sync	Packet loss, DBD retransmits, unstable link
Loading	Missing LSAs	LSR/LSU loss or bad/corrupt LSA
Full	Adjacency complete	Then check LSDB, SPF, route filters

NORMAL STATES

- Down on a shutdown/passive interface.
- 2-Way between DROTHER routers.
- Brief Exchange/Loading during convergence.

STUCK STATES

- Init for one-way Hello visibility.
- ExStart/Exchange for MTU or duplicate router ID.
- Loading for LSA loss or LSDB corruption.

KEY POINT * Full is the adjacency goal, but 2-Way can be correct on broadcast/NBMA when neither router is DR or BDR.

MUST MATCH

- Area ID
- Hello/dead timers
- Authentication
- Stub/NSSA flag
- MTU during DBD exchange
- Network mask on broadcast/NBMA

MUST BE UNIQUE

- Router ID in the OSPF domain
- Interface IP addressing
- LSA IDs per originating router

MUST BE REACHABLE

- Interface line protocol up
- Protocol 89 allowed
- Multicast 224.0.0.5/6 allowed
- NBMA neighbor statement if required

MAY DIFFER

- Cost
- Priority
- Reference bandwidth
- Passive status on unrelated interfaces

VERIFY REQUIREMENTS

```
show ip ospf interface
show ip ospf neighbor detail
show ip protocols
show run interface <interface>
debug ip ospf hello
```

CLASSIC MISREAD

MTU mismatch usually appears in ExStart or Exchange, not Down. If the neighbor never starts, check basic Hello visibility first.

WHEN ONLY ONE SIDE SHOWS A NEIGHBOR

- Check unidirectional ACLs.
- Confirm source interface/subnet.
- Verify multicast reaches both directions.

WHEN BOTH SIDES SEE EACH OTHER

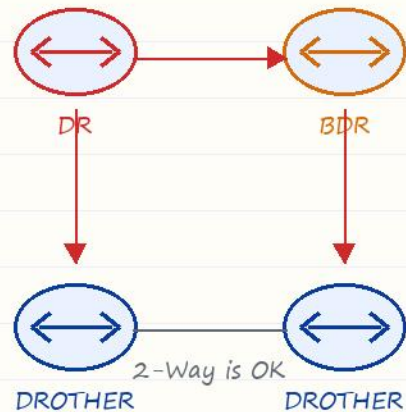
- Compare timers and options.
- Check state transition and DBD phase.
- Move next to LSDB and route checks.

KEY POINT ✨ The cleanest OSPF check is boring: same area, same timers, same auth, compatible area type, and no filter blocking protocol 89.

Type	Typical use	DR/BDR?	Operational note
Broadcast	Ethernet	Yes	224.0.0.5/6 multicast
Point-to-point	Serial, p2p VLAN	No	Always forms Full with peer
NBMA	Frame Relay style	Yes	Needs neighbor statements
Point-to-multipoint	Hub/spoke variants	No	Models spokes as p2p links

DR / BDR RULES

- Highest OSPF priority wins; tie-breaker is highest router ID.
- Priority 0 means never become DR or BDR.
- Election is not preemptive; new higher priority does not replace current DR.
- DROTHER routers normally stay 2-Way with each other.
- All routers form Full adjacency with DR and BDR.



ELECTION GOTCHAS

- Changing priority does not preempt current DR.
- Clear/reload may be needed to force a new election.
- The DR's router ID may not be the interface IP.

NETWORK-TYPE GOTCHAS

- Broadcast vs point-to-point mismatch can create odd states.
- NBMA needs manual neighbors.
- Point-to-multipoint avoids DR logic.

KEY POINT ✨ On shared media, OSPF reduces adjacency scale by using a DR/BDR. Do not chase DROTHER-to-DROTHER 2-Way as a failure.

Type	Name	Originator	Scope	Meaning
Type 1	Router LSA	Every router	Inside one area	Lists router links and costs
Type 2	Network LSA	DR	Inside one area	Lists routers on broadcast/NBMA segment
Type 3	Summary LSA	ABR	Between areas	Advertises inter-area prefixes

HOW TO READ THE LSDB

- Type 1 and Type 2 describe topology inside the area.
- Type 3 is how an ABR tells another area about a prefix.
- If all routers in an area disagree on Type 1/2, the area is unhealthy.
- If inter-area routes are missing, inspect Type 3 LSAs and ABR behavior.

COMMANDS

- show ip ospf database router
- show ip ospf database network
- show ip ospf database summary
- show ip ospf database <lsa-id>
- show ip route ospf

TYPE 1 CLUES

- Missing link in Router LSA means topology is incomplete.
- Wrong cost here changes SPF for the whole area.
- A changed router ID creates new LSAs.

TYPE 2 / 3 CLUES

- Network LSA exists only when there is a DR.
- Summary LSA proves what the ABR advertises.
- Missing Type 3 usually means ABR or filtering.

KEY POINT ✨ Type 1 + Type 2 build the intra-area map. Type 3 is the ABR carrying reachability between areas.

Type	Name	Originator	Advertises	Operational note
Type 4	ASBR Summary	ABR	Route to ASBR	Needed so other areas know how to reach the ASBR
Type 5	AS External	ASBR	External route	Redistributed route, blocked from stub areas
Type 7	NSSA External	ASBR in NSSA	NSSA external	Translated to Type 5 by the NSSA ABR

EXTERNAL ROUTE TYPES

- E1/N1 adds internal cost to reach the ASBR plus external cost.
- E2/N2 compares external metric first; internal cost is tie-breaker.
- Cisco default for redistributed external routes is E2.
- Forwarding address can change the next-hop behavior for external LSAs.

COMMON MISREADS

- Type 4 is not the external prefix. It is reachability to the ASBR.
- Type 5 missing in a stub area may be correct by design.
- Type 7 lives inside NSSA until translated by the ABR.
- Full adjacency does not guarantee external LSAs are allowed in the area.

E1 / N1 USE CASE

- Use when internal path to the ASBR should influence best path.
- Better for multi-exit redistribution designs.
- Total metric reflects external plus internal cost.

E2 / N2 USE CASE

- Default for many Cisco redistributions.
- External metric dominates path choice.
- Good when all exits are meant to be treated similarly.

KEY POINT ✨ When redistributed routes are missing, check area type first, then Type 4/5/7 LSAs, then route policy.



AREA DESIGN

- Area 0 is the backbone.
- All non-backbone areas must attach to Area 0.
- ABR has interfaces in multiple areas.
- ABR advertises Type 3 summary LSAs between areas.
- Virtual links are repair tools, not a default design.

ROUTER ROLES

- Internal router: all OSPF interfaces in one area.
- Backbone router: at least one interface in Area 0.
- ABR: connects Area 0 to another area.
- ASBR: redistributes routes from another protocol/source.
- A router can be both ABR and ASBR.

CHECK BACKBONE HEALTH

```
show ip ospf border-routers
show ip ospf database summary
show ip route ospf
```

ABR CHECKS

- Does the ABR have Area 0 adjacency?
- Does it originate expected Type 3 LSAs?
- Is summarization hiding detail intentionally?

ASBR CHECKS

- Is redistribution active?
- Does Type 5 or Type 7 exist?
- Is a Type 4 route to the ASBR present where needed?

KEY POINT ✨ Most multi-area problems are either broken Area 0 reachability, wrong area type, or ABR filtering/summarization.

Area type	Type 3/4	Type 5	Type 7	Notes
Standard	Yes	Yes	No	Full LSA visibility
Stub	Yes	No	No	Default route from ABR replaces externals
Totally Stubby	No	No	No	Only default route for inter-area/external
NSSA	Yes	No	Yes	Local externals allowed as Type 7
Totally NSSA	No	No	Yes	Type 7 local external + default route

MATCH ON ALL ROUTERS

- Stub/NSSA settings must match between neighbors.
- Mismatch can prevent adjacency formation.
- Area type changes route visibility by design.
- Do not treat missing Type 5 LSAs in stub/NSSA as a failure.

FIELD CHECK

- show ip ospf
- show ip ospf database external
- show ip ospf database nssa-external
- show ip route ospf
- show run | section router ospf

STUB DESIGN FIT

- Good for simple leaf areas.
- Blocks external LSA flooding.
- ABR injects a default route.

NSSA DESIGN FIT

- Good when a leaf area must redistribute local routes.
- Uses Type 7 inside the NSSA.
- ABR translates Type 7 to Type 5.

KEY POINT ✨ Area type is policy. Before chasing a missing external route, ask whether the area is supposed to see that LSA type.

COST FORMULA

$\text{cost} = \text{reference bandwidth} / \text{interface bandwidth}$

- Cisco default reference bandwidth is historically 100 Mbps.
- Set reference-bandwidth consistently on every OSPF router.
- Higher bandwidth means lower OSPF cost.
- Manual interface cost overrides calculated cost.

PATH PREFERENCE

Order	Path type	Route code
1	Intra-area	O
2	Inter-area	O IA
3	External Type 1	E1 / N1
4	External Type 2	E2 / N2

ECMP

- Equal-cost paths can install together.
- Unequal-cost load sharing is not native OSPF.
- Check CEF/FIB to prove actual forwarding buckets.
- Different vendors/platforms vary in max ECMP paths.

METRIC COMMANDS

```
show ip ospf interface
show ip route <prefix>
show ip cef <prefix> detail
auto-cost reference-bandwidth <mbps>
```

PATH SELECTION CHECKS

- Compare route type before comparing metric.
- Check total cost along every hop.
- Use traceroute/CEF to verify the installed path.

COMMON METRIC MISTAKES

- Leaving default reference bandwidth in high-speed cores.
- Manually overriding one link and forgetting the peer side.
- Assuming OSPF does unequal-cost load sharing.

KEY POINT ✨ Use a consistent reference bandwidth. Otherwise 1G, 10G, and faster links can all collapse to the same OSPF cost.

ABR SUMMARY

- Summarize inter-area routes with area range.
- Occurs between areas.
- Reduces Type 3 LSA detail.

ASBR SUMMARY

- Summarize redistributed routes with summary-address.
- Affects external Type 5/7 routes.
- Useful at redistribution boundaries.

FILTERING

- Distribute-list may filter RIB installation.
- Area filter-list controls Type 3 summaries.
- Route-maps control redistribution.

DEFAULT ROUTE

- default-information originate injects default.
- Always keyword injects even without local default.
- NSSA/stub defaults are area behavior.

COMMON CONFIG CLUES

```
area 10 range 10.10.0.0 255.255.0.0
summary-address 172.16.0.0 255.255.0.0
area 10 filter-list prefix BLOCKED in
default-information originate
redistribute connected subnets route-map RM-OSPF
```

TROUBLESHOOTING NOTE

If the LSA exists but the route is missing from the RIB, look for administrative distance, better route source, filtering, or unresolved forwarding address.

WHEN THE LSA IS MISSING

- Check ABR/ASBR origination.
- Check area type and flooding scope.
- Check filters at the area boundary.

WHEN THE LSA EXISTS

- Check route installation.
- Check better AD/source route.
- Check next-hop/forwarding address resolution.

KEY POINT ✨ Summarization changes what is advertised. Filtering changes what is accepted, flooded, redistributed, or installed. Check the exact control point.

Topic	OSPFv2	OSPFv3
Transport	OSPFv2 rides directly in IPv4	OSPFv3 rides directly in IPv6
Neighbors	IPv4 address on interface	IPv6 link-local next hop
Addressing	IPv4 prefixes in LSAs	IPv6 prefixes separated from topology LSAs
Instances	Process/area/interface	Multiple instances per link supported
Auth	Native OSPFv2 auth	Typically IPsec/AH or platform AF auth

CISCO-LIKE CHECKS

```
show ospfv3 neighbor
show ospfv3 interface
show ipv6 route ospf
show ospfv3 database
show ipv6 ospf interface
```

KEY FIELD POINTS

- Link-local addresses matter for neighbor next hop.
- IPv6 multicast: ff02::5 and ff02::6.
- Router ID is still 32-bit, even in IPv6-only networks.
- OSPFv3 can support multiple address families on some platforms.

OSPFv3 GOTCHAS

- No IPv4 address is required on the link for IPv6 adjacency.
- Router ID still must be configured or derived.
- ACLs must permit IPv6 OSPF, not just IPv4 protocol 89.

JUNOS CHECKS

- show ospf3 neighbor
- show ospf3 interface
- show route protocol ospf3
- show ospf3 database

KEY POINT ✨ OSPFv3 feels familiar, but the link-local next hop and 32-bit router ID are the two details that catch people in the field.

AUTHENTICATION

- Use authentication on untrusted/shared segments.
- Mismatch blocks adjacency.
- Key rollover should be planned, not improvised.

TIMERS

- Hello/dead must match.
- Throttle SPF/LSA timers prevent churn.
- Aggressive timers without BFD can flap under load.

BFD

- Fast failure detection below OSPF.
- Both sides must support and agree.
- Watch CPU and control-plane policing.

GRACEFUL RESTART

- Preserves forwarding during planned control-plane restart.
- Helper mode behavior is vendor specific.
- Does not fix a broken data plane.

LSA CONTROL

- Demand circuits reduce periodic refresh in special designs.
- LSA pacing protects CPU.
- Max-metric on startup avoids blackholing during warmup.

PRODUCTION NOTE

- Avoid broad debug on busy routers.
- Prefer targeted neighbor/interface checks.
- Capture timestamps before clearing state.

STABILITY COMMANDS

- `show ip ospf statistics`
- `show ip ospf events`
- `show logging | include OSPF`
- `show bfd neighbors detail`

BEFORE CHANGING TIMERS

- Know link latency and CPU headroom.
- Use BFD for fast failure when possible.
- Apply values consistently, then monitor churn.

KEY POINT ✳ Fast detection is only useful if the network can reconverge cleanly. Pair timers, BFD, SPF throttle, and operational guardrails.

TROUBLESHOOTING WORKFLOW

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Interface	Hello	State	LSDB	SPF/RIB	FIB/Data
link/IP/passive/ACL	area/timers/auth/stub flag	Down -> Full transition	missing or wrong LSAs	route installed?	CEF, MTU, ACL, next hop

Symptom	Check first	Best command
No neighbor	Interface, passive, ACL, protocol 89, area	show ip ospf int brief
Stuck Init	One-way Hello, timer mismatch, ACL	show ip ospf nei detail
Stuck 2-Way	Normal DROTHER or network-type issue	show ip ospf interface
Stuck ExStart	MTU mismatch, duplicate router ID	show ip ospf int
Stuck Loading	LSA request/loss, corrupt LSA	show ip ospf database
Full no route	Area type, filter, summary, better AD	show ip route ospf
Route no traffic	CEF/FIB, MTU, ACL, next hop	show ip cef detail

CAPTURE EVIDENCE FIRST

- Neighbor detail output.
- Interface OSPF details.
- Database entry for missing prefix.
- Route table and CEF entry.

THEN CHANGE ONE THING

- Fix the most likely mismatch.
- Wait for convergence.
- Verify neighbor, LSDB, RIB, and FIB again.

KEY POINT ✨ Do not clear the process as a diagnostic reflex. Capture state first, or you destroy the evidence.

RUN IN THIS ORDER

```
show ip ospf neighbor
show ip ospf neighbor detail
show ip ospf interface brief
show ip ospf interface <int>
show ip ospf database
show ip route ospf
show ip protocols
show ip ospf border-routers
debug ip ospf adj
debug ip ospf hello
```

COMMON FIELD SCENARIOS

- New link comes up but no neighbor: passive-interface or ACL.
- Neighbor stuck ExStart after circuit migration: MTU mismatch.
- Full adjacency but remote subnet gone: stub/NSSA or ABR filter.
- DR changed unexpectedly after reload: election is not preemptive.
- External route missing in other area: inspect Type 4 plus Type 5/7.
- IPv6 neighbor up but route odd: check link-local next hop.

DEBUG SAFETY

- Use terminal monitor carefully, and turn debug off immediately after capture.
- Prefer neighbor/interface scoped checks first.
- In production, capture show outputs before clearing neighbors or process.
- If using BFD, troubleshoot BFD session state alongside OSPF state.

QUICK ROOT CAUSE MAP

- Down: no Hello visibility.
- Init: one-way communication.
- ExStart: MTU or router ID.
- Full/no route: LSDB, area, filter, or RIB.

ESCALATION PACKAGE

- Topology snippet.
- Before/after show commands.
- Exact timestamps.
- Recent config/circuit changes.

KEY POINT ✨ The most common OSPF fix is not advanced: correct the area/timer/auth/MTU mismatch, then let the protocol converge.

CISCO IOS / IOS-XE

```
router ospf 10
router-id 1.1.1.1
passive-interface default
no passive-interface Gi0/0
network 10.0.0.0 0.0.0.255 area 0
auto-cost reference-bandwidth 100000
interface Gi0/0
ip ospf 10 area 0
ip ospf cost 10
```

CISCO IOS-XR

```
router ospf CORE
router-id 1.1.1.1
area 0
interface Loopback0
interface TenGigE0/0/0/0
network point-to-point
!
show ospf neighbor
show ospf database
```

JUNIPER JUNOS

```
set protocols ospf area 0 interface lo0.0 passive
set protocols ospf area 0 interface ge-0/0/0.0
set protocols ospf area 0 interface ge-0/0/0.0 metric 10
set protocols ospf reference-bandwidth 100g
show ospf neighbor
show ospf database
show route protocol ospf
```

CONFIG REVIEW CHECKLIST

- Router ID pinned and unique.
- Reference bandwidth consistent domain-wide.
- Passive default used intentionally.
- Area type matches all neighbors.
- Summaries documented at ABR/ASBR.
- Authentication and key rollover documented.

AREA EXAMPLES

- area 0 = backbone transit.
- area 10 = access/campus block.
- area 100 stub = simple leaf.
- area 200 nssa = leaf with local redistribution.

CHANGE-CONTROL NOTES

- Changing area type resets adjacency.
- Changing router ID rebuilds LSAs.
- Changing reference bandwidth can move traffic.
- Summaries can hide specific failures.

KEY POINT ✨ Favor interface-level OSPF config for clarity during audits. It is harder to misread than broad network statements.



Try: break MTU, area type, passive-interface, then troubleshoot from show outputs.

Focus	Command	Question
Neighbor	show ip ospf neighbor	Is state Full or expected 2-Way?
Interface	show ip ospf interface	Area, timers, MTU, type, cost
Database	show ip ospf database	Are LSAs present and sane?
Routes	show ip route ospf	Did SPF result install?
ABR/ASBR	show ip ospf border-routers	Can area reach the border?
Forwarding	show ip cef <prefix> detail	Does FIB match RIB?

BEST PRACTICES

- Use stable loopbacks for router IDs.
- Document area boundaries and summary points.
- Keep reference bandwidth consistent.
- Use BFD deliberately where fast failure detection matters.
- Know which 2-Way states are normal.

GOLDEN RULE

OSPF troubleshooting is evidence work: neighbor state tells you where to start; the LSDB and RIB tell you why the route did or did not install.

KEY POINT ✨ When in doubt, follow the chain: Hello -> Neighbor -> LSDB -> SPF -> RIB -> FIB.